

Talofibular Ligament Injury

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Overview

Background

Ligamentous injuries of the ankle are common among athletes.[1] Inversion injuries of the ankle account for 40% of all athletic injuries. The anterior talofibular ligament (ATFL) and the calcaneofibular ligament (CFL) are sequentially the most commonly injured ligaments when a plantar-flexed foot is forcefully inverted. The posterior talofibular ligament (PTFL) is rarely injured, except in association with a complete dislocation of the talus.[2,3,4]

Ligamentous injuries of the ankle are classified into the following three categories, depending on the extent of damage to the ligaments[5]:

- Grade I is an injury without macroscopic tears - No mechanical instability is noted; pain and tenderness are minimal
- Grade II is a partial tear - Moderate pain and tenderness are present; mild-to-moderate joint instability may be present
- Grade III is a complete tear - Severe pain and tenderness, inability to bear weight, and significant joint instability are noted

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Epidemiology

Lateral ankle sprains make up over 75% of all ankle sprains, and of that portion, about 73% are ATFL injuries.[1]

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Functional Anatomy

The lateral articular capsule of the ankle can be divided into anterior and posterior segments. The anterior segment attaches proximally to the anterior portion of the distal tibia superior to the articular surface and to the border of the articular surface of the medial malleolus. The posterior segment attaches distally to the talus just posterior to its superior articular facet and attaches laterally to the depression in the medial surface of the lateral malleolus.[2,3,4]

The ATFL is intracapsular and attaches anteriorly to the anterior border of the distal fibula and laterally to the neck of the talus. The PTFL attaches posteriorly to the digital fossa of the fibula and laterally to the lateral tubercle on the posterior portion of the talus.

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Sport-Specific Biomechanics

The talofibular ligaments along with the CFL are components of the lateral ligament complex. This complex becomes stressed when the ankle is inverted and plantar flexed.[6] Supination of the foot in neutral flexion usually results in injury of the CFL. Supination and adduction injuries tear both the ATFL and the CFL.

The PTFL is the strongest of the lateral ligaments, and extreme inversion with plantar flexion is required to place the PTFL under stress; as a result, the PTFL is less commonly injured.[6] Transient subluxation or dislocation of the talus from the tibial mortise usually results in injury to all three lateral ligaments. Prevention of anterior displacement of the talus is primarily a function of the

ATFL. Little additional motion occurs when the CFL also is damaged. Instability to inversion is greater when both the CFL and the ATFL are injured than when either ligament is injured alone.



Prognosis

Eighty percent of patients with lateral ankle injuries make a full recovery following conservative rehabilitation.

Up to 20% of patients with an acute inversion injury develop chronic functional instability. Electromyography (EMG) has demonstrated prolonged reaction times of the peroneal muscle in this group of patients. Strengthening and proprioception exercises can lead to improvement. Patients whose injury does not respond and who have continued mechanical laxity and functional instability may be candidates for lateral ligament reconstruction.



Patient Education

To help prevent recurrent injury, patients should be instructed regarding the proper techniques for ankle taping and bracing. Ankle-strengthening and proprioception exercises should also be an important part of rehabilitation, and the patient should be instructed in an appropriate home exercise program.

For excellent patient education resources, visit eMedicineHealth's First Aid and Emergencies Center.



Presentation

History

The history portion of the examination for a suspected talofibular ligament injury should include the following:

- Mechanism of injury
- Time of injury
- Concurrent injuries
- Position of the body at the time of injury
- Rotational component to injury
- Ability or inability to bear weight immediately after the injury
- Time of onset of pain and swelling (immediate or delayed)
- Whether the patient heard or felt a popping sound or sensation at the time of the injury
- Information regarding any previous ankle injuries



Physical Examination

The physical examination for a suspected talofibular injury should include the following:

- Inspect the ankles - Both ankles should be completely uncovered so the injured side can be compared with the uninjured side. Note any swelling, ecchymosis, lacerations, abrasions, or deformities.
- Palpate the injured ankle - Note any tenderness or crepitus.
- Test the range of motion - Patients with ligamentous injuries, especially to the anterior talofibular ligament (ATFL), will have limited and painful inversion of their ankle.

- Perform a neurovascular examination of the foot distal to the injury - Document the findings.
- Assess the stability of the ankle joint - The anterior drawer test assesses the stability of the lateral ligaments. To perform this test, the foot is placed in slight inversion and 20° of plantar flexion. The heel is grasped firmly and drawn forward by the examiner while the tibia is stabilized by the examiner's other hand. A positive sign occurs when the talus moves forward on the tibia. The injured side should also be tested for maximal inversion compared with the uninjured side. If the ATFL is torn, forward motion is detected on performing the anterior drawer test. If the ATFL and the CFL are torn, abnormal inversion is elicited.
- Assess the stability of the calcaneofibular ligament - Grade I sprains are stable to stress testing. Grade II sprains have a mildly increased anterior drawer test and are stable to inversion. Grade III sprains are unstable to both the anterior drawer test and the talar tilt test. Instability with these tests indicates a complete tear of the ATFL and at least a partial tear of the CFL.
- Perform a neurologic exam - This should include testing patients' balance. Have them stand on their uninjured foot, initially with their eyes open; then, have them close their eyes. Then have patients do this with the injured foot and compare. Ankle injuries will often disrupt the nerves, causing patients to have poor balance.



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Diagnostic Considerations

Along with those listed in the next section, conditions to consider in the differential diagnosis of talofibular ligament injury include the following:

- Achilles tendonitis
- Ankle impingement syndrome
- Calcaneofibular ligament injury

Differential Diagnoses

- [Achilles Tendon Injuries](#)
- [Ankle Fracture Management in the ED](#)
- [Ankle Sprain](#)
- [Peroneal Tendon Syndromes](#)



Workup

Laboratory and Imaging Studies

Laboratory studies

Laboratory studies are not indicated in the workup of ligamentous injuries of the ankle.

Imaging studies

Indications for imaging studies in cases of suspected talofibular ligament injuries include the following:

- Bony tenderness or deformity
- Suspicion of a fracture or syndesmotic injury

- Severe pain or swelling that makes the physical examination unreliable
- Inability to walk

Initial radiologic studies of the ankle should include the following:

- An anteroposterior (AP) view with the ankle in slight adduction
- A true lateral view
- A mortise view (45° oblique view with the ankle in dorsiflexion)
- Consider stress views of the ankle
- If a syndesmotic injury is suspected, then AP and lateral views of the tibia and fibula should also be obtained to rule out associated fibular fractures

Magnetic resonance imaging (MRI) may be useful in evaluating the soft-tissue anatomy of the ankle, such as ligaments and tendons.[7,8] This imaging modality is not typically an initial test performed, but MRI may be useful in the patient who is not healing, in whom a stress fracture is suspected, or who has chronic ankle pain and instability.

A study by Xu et al showed that for diagnosis of partial tears of the anterior talofibular ligament (ATFL), the sensitivity of three-dimensional (3-D) MRI was 78%, and the specificity was 100%.[9]

A study with 25 participants reported that stress ultrasonography provides a safe, repeatable, and quantifiable method of measuring the talofibular interval and may augment manual stress examinations in acute ankle injuries.[10]

A systematic review and meta-analysis by Cao et al found that for acute ATFL injuries, the pooled sensitivity rates of MRI and ultrasonography were 82.1% and 88.6%, respectively. The pooled specificity rates of MRI and ultrasonography were 37.8% and 90.3%, respectively.[11]

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Treatment

Acute Phase

Rehabilitation program

Physical therapy

Initial treatment of all grades of lateral ankle sprains consists of rest, ice, compression, and elevation (RICE) as well as nonsteroidal anti-inflammatory drugs (NSAIDs).[8,12,13] Ice should be applied to the injured ankle for approximately 20 minutes, three to four times per day. Compressive dressings should be used to control swelling. Weight bearing should be encouraged as soon as it is tolerated.

With grade III injuries, an ankle brace should be worn at all times until the patient can ambulate pain free. Ankle braces can be used for support in all ankle sprains and may help to promote earlier ambulation. Studies have found an air stirrup brace and ACE wrap are beneficial in grade 1 ankle sprains and may help in grade II and III sprains as well.[8]

The early phase of rehabilitation is begun approximately 48 hours post injury.[14] Icing is continued and range of motion exercises are initiated. Writing the alphabet with the great toe moves the ankle through full range of motion in all planes. Stationary biking and stretching of the Achilles tendon are also beneficial.

As strength and mobility improve, isometric exercises for ankle dorsiflexion, plantar flexion, inversion, and eversion are initiated. The isometric exercises are followed by resistance exercises (initially using a Thera-Band strap) and then heel and toe raises. Agility training also aids in returning the athlete to sports. In addition, proprioceptive and balance training are extremely important to help the patient heal fully, especially in jumping athletes. Proprioceptive training should be incorporated into all rehabilitation protocols.

In grade III ankle sprains, some studies have shown that early mobilization and rehabilitation may provide earlier functional recovery relative to surgery, and there is general agreement to try a 6-week period of conservative management, including early, controlled mobilization and rehabilitation, before considering surgery.[14,15]

Also, no difference is found in long-term outcome when comparing early surgical repair with delayed surgical repair following failed conservative therapy.[15] Therefore, there is no indication for routine early surgical repair.

Surgical intervention

Primary repair of acute lateral ligament tears is rarely indicated. Open repair seems to offer no advantage over closed management at the time of the initial injury. Delayed repair may be necessary in patients with chronic mechanical instability on clinical examination and functional instability; however, surgical intervention in these cases should be considered only after an aggressive rehabilitation program has been unsuccessful.

A randomized, controlled trial by Vuurberg et al found that in anterior talofibular ligament (ATFL) repair for patients with chronic ankle instability, functional improvement from arthroscopic surgery (20 patients) was equal to that from open repair (21 patients). At none of the follow-up intervals did any of the subscales of the Foot and Ankle Outcome Score (FAOS) show a statistically significant difference between the two groups.[16]

A retrospective cohort study by Nakasa et al of 60 patients (65 ankles) with chronic lateral ankle instability who were treated with arthroscopic ATFL repair reported that a preoperative finding of concomitant calcaneofibular ligament (CFL) injury on magnetic resonance imaging (MRI) increases the risk of recurrent, postoperative instability. This was particularly the case when the injured CFL had a thin or wavy morphology.[17]

Consultations

An emergent consultation is rarely required.

Consultation with an orthopedic surgeon should be obtained for patients with unstable ankles, dislocations, or associated fractures.

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Recovery Phase

Rehabilitation program

Physical therapy

When the early phase rehabilitation goals of decreased swelling, full weight bearing, and no tenderness to palpation are met, more aggressive strengthening and proprioceptive training are added. Increased stretching of the Achilles tendon, as well as the gastrocnemius and soleus muscles, is performed using an incline board. Thera-Band exercises are continued for strengthening. Exercises such as one-leg stands and wobble board training are added for proprioception. This training continues until the ankle is at 80-90% of full strength and there are no deficits in proprioception. When these goals are met, the patient may be discharged from therapy.

Surgical intervention

Maximum benefit from conservative therapy is reached after approximately 10 weeks of active rehabilitation. At this time, 20% of athletes continue to have symptoms secondary to either a functional or mechanical instability. If the patient has reached his or her maximum benefit from functional rehabilitation and has a persistent deficit, then surgical reconstruction should be considered.[15,18,19]

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Maintenance Phase

Rehabilitation program

Physical therapy

The patient should be independent with a home exercise program featuring sport-specific activities and should gradually return to play when the functional goals are met. The physician and/or physical therapist should recommend taping or bracing the ankle upon returning to activity. Taping or bracing a previously injured ankle during athletic activity has been shown to reduce injury recurrence.

Return to play

Athletes may return to sports following a talofibular ligament injury when they are able to run and pivot without pain while the ankle is braced. Bracing and taping of the injured ankle is recommended for approximately 6 weeks. Some athletes may feel more comfortable bracing or taping for several months.

Prevention

Completion of an adequate rehabilitation program, as well as functional bracing or taping for 6 months following the injury, minimizes the chance of recurrent injury.

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Medication

Analgesics, Opioid Combos

Codeine/acetaminophen (Tylenol #3, Tylenol #4, Tylenol with Codeine)

Hydrocodone/acetaminophen (Anexsia, Hycet, Lorcet)

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Analgesics, Other

Acetaminophen (Tylenol, Little Fevers Children's Fever/Pain Reliever, Little Fevers Infant Fever/Pain Reliever)

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NSAIDs

Ibuprofen (Advil, Motrin, PediaCare Children's Pain Reliever/Fever Reducer IB)

Naproxen (Aleve, Anaprox, Anaprox DS)

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